

Reference	Qty	Value	Footprint	Description	LCSC Part Number	MPN	Link	Price per single unit	Price per Qty
C2,C10,C18	3	1n	Capacitor_SMD:C_0603	Unpolarized cape	C163508	C0603C102J3GCAUTO	https://www.lcsc.com	\$0.0127	\$0.0381
C4,C7	2	10u	Capacitor_SMD:C_1206	Unpolarized caps	C105175	MSASU31LBB5106KTNA01	https://www.lcsc.com	\$0.4207	\$0.8414
C5	1	220u	Capacitor_SMD:Cp_Elec	Polarized capacit	C278406	EEFFP1V221AP	https://www.lcsc.com	\$0.4923	\$0.4923
C6,C12,C14,C16,C17,C20,C25,C26,C27,C28	10	100n	Capacitor_SMD:C_0603	Unpolarized caps	C346190	CL10B104KA8WPNC	https://www.lcsc.com	\$0.0278	\$0.2780
C8,C9,C11,C21,C22,C23,C24	7	1u	Capacitor_SMD:C_0603	Unpolarized caps	C85862	GCM188R71E105KA64D	https://www.lcsc.com	\$0.0893	\$0.6251
C13,C15	2	22u	Capacitor_SMD:C_0603	Unpolarized caps	C2762594	CL10A226M07JZNC	https://www.lcsc.com	\$0.1292	\$0.2584
C19	1	4.7u 10V	Capacitor_Tantalum_SN	Polarized capacit	C7187	TAJA475K016RNJ	https://www.lcsc.com	\$0.2206	\$0.2206
D1,D2	2	BAV199.215	Package_TO_SOT_SMD	BAV99 High-spec	C40919	BAV199.215	https://www.lcsc.com	\$0.0660	\$0.1320
D3	1	ESDA6V1SC5	Package_TO_SOT_SMD	Quad TVS Diode	C2935146	ESDA6V1SC5	https://www.lcsc.com	\$0.2890	\$0.2890
D4,D5	2	SMAJ24CA	Diode_SMD:D_SMA		C148223	SMAJ24CA	https://www.lcsc.com	\$0.1069	\$0.2138
D6	1	PMEG6010CEH.115	Diode_SMD:D_SOD-123	Schottky diode, s	C53113897	PMEG6010CEH.115-JSM	https://www.lcsc.com	\$0.0469	\$0.0469
D7,D8	2	BAT54SW	Package_TO_SOT_SMD	Vr 30V, If 200mA,	C78606	BAT54SW.115	https://www.lcsc.com	\$0.0535	\$0.1070
D9	1	BZV55-B12.115	Diode_SMD:D_MinimELU	Zener diode, sma	C426807	BZV55-B12.115	https://www.lcsc.com	\$0.0979	\$0.0979
D10,D11	2	1N4148WSF-7	Diode_SMD:D_SOD-323		C119617	1N4148WSF-7	https://www.lcsc.com	\$0.6677	\$1.3354
D12	1	WS2812B-2020	Footprints:RGB_LED_WS		C52917434	WS2812B-2020-V6	https://www.lcsc.com	\$0.0916	\$0.0916
F1	1	7A	Footprints:0154100DR	Fuse	C206918	0154007.DR	https://www.lcsc.com	\$1.9183	\$1.9183
FB1	1	BLM18AG331SN1D	Inductor_SMD:L_0603_1	Ferrite bead, sma	C113010	BLM18AG331SN1D	https://www.lcsc.com	\$0.0139	\$0.0139
J1	1	Screw_Terminal_01x0	Footprints:TerminalBloc	Generic screw ter					\$0.0000
J2	1	USB4105-GF-A-060	Connector_USB:USB_C_	USB 2.0-only 16P	C3025063	USB4105-GF-A-060	https://www.lcsc.com	\$1.1952	\$1.1952
J3,J4	2	Conn_01x01_Pin	Connector_PinHeader_2	Generic connect	C7501259	ZX-P22.54-1-1PZZ	https://www.lcsc.com	\$0.0146	\$0.0292
J5	1	Conn_01x05_Socket	Connector_PinSocket_2	Generic connect	C7171947	215298-5	https://www.lcsc.com	\$0.8446	\$0.8446
L1	1	47uH	Inductor_SMD:L_APV_A	Inductor	C706600	SMMS0630-470MT	https://www.lcsc.com	\$0.1268	\$0.1268
Q2,Q4	2	AO4435	Package_SO:SOIC-8_3_9	6.7A Id, -20V Vd:	C3000	IRF7416TRPBF	https://www.lcsc.com	\$0.4558	\$0.9116
Q3,Q5	2	BSR14	Package_TO_SOT_SMD	TRANS NPN 40V	C258201	BSR14	https://www.lcsc.com	\$0.2776	\$0.5552
R1,R2,R9,R10,R19,R31,R34	7	1.5k	Resistor_SMD:R_0603_1	Resistor, small s	C2930050	FRC0603F1501TS	https://www.lcsc.com	\$0.0027	\$0.0189
R4	1	1M	Resistor_SMD:R_0603_1	Resistor, small s	C2907003	FRC0603F1004TS	https://www.lcsc.com	\$0.0027	\$0.0027
R5,R6	2	390k	Resistor_SMD:R_0603_1	Resistor, small s	C2907158	FRC0603J394 TS	https://www.lcsc.com	\$0.0022	\$0.0044
R7,R16	2	10R	Resistor_SMD:R_0603_1	Resistor, small s	C2907091	FRC0603J100 TS	https://www.lcsc.com	\$0.0027	\$0.0054
R8,R28,R30	3	10k	Resistor_SMD:R_0603_1	Resistor, small s	C4134045	RCS060310K0JNEA	https://www.lcsc.com	\$0.0476	\$0.1428
R17,R20,R35	3	20k	Resistor_SMD:R_0603_1	Resistor, small s	C2907011	FRC0603F2002TS	https://www.lcsc.com	\$0.0027	\$0.0081
R11	1	RMF2512FTR005	Resistor_SMD:R_2512_6	Resistor, small s	C48533600	FRM253WJR005TM	https://www.lcsc.com	\$0.0335	\$0.0335
R12,R13,R24,R26,R29,R32,R37,R38,R39	9	5k1	Resistor_SMD:R_0603_1	Resistor, small s	C2907044	FRC0603F5101TS	https://www.lcsc.com	\$0.0028	\$0.0252
R14,R15	2	240k	Resistor_SMD:R_0603_1	Resistor, small s	C2930036	FRC0603J244TS	https://www.lcsc.com	\$0.0023	\$0.0046
R18,R23,R25	3	100R	Resistor_SMD:R_0603_1	Resistor, small s	C2076611	CRCW0603100RJNEAC	https://www.lcsc.com	\$0.0156	\$0.0468
R20,R22,R27	3	130k	Resistor_SMD:R_0603_1	Resistor, small s	C2907099	FRC0603J134 TS	https://www.lcsc.com	\$0.0024	\$0.0072
R21	1	27k	Resistor_SMD:R_0603_1	Resistor, small s	C2930084	FRC0603F2702TS	https://www.lcsc.com	\$0.0028	\$0.0028
SW1,SW2	2	SW_Omron_B3FS	Button_Switch_SMD:SW	Omron B3FS 6x6	C5594007	B3FS-1050	https://www.lcsc.com	\$1.2121	\$2.4242
U1	1	AD8629ARZ	Package_SO:SOIC-8_3_9	Dual Precision, V	C23487	AD8629ARZ-REEL7	https://www.lcsc.com	\$4.9347	\$4.9347
U3	1	AP63300UW-7	Package_TO_SOT_SMD	2A, 500kHz Buck	C2158012	AP63300UW-7	https://www.lcsc.com	\$0.6580	\$0.6580
U4	1	STUSB4500QTR	Footprints:QFN-24-1EP	Stand-alone USB	C2678061	STUSB4500QTR	https://www.lcsc.com	\$2.1956	\$2.1956
U5	1	MIC5219-3.3YM5	Package_TO_SOT_SMD	500mA low drop	C29613	MIC5219-3.3YM5-TR	https://www.lcsc.com	\$0.5892	\$0.5892
U6	1	OLED0.91_128x32_I2	Footprints:OLED_128x3	0.91 128x32 SSD	C5248081	HS91L02W2C01	https://www.lcsc.com	\$2.2378	\$2.2378
U7	1	STM32F103C8T6	Package_QFP:LQFP-48_ST	Microelectroni	C8734	STM32F103C8T6	https://www.lcsc.com	\$2.0073	\$2.0073
U8	1	MCP9700AT-ETT	Package_TO_SOT_SMD	Low power, analc	C127949	MCP9700AT-E/TT	https://www.lcsc.com	\$0.4460	\$0.4460
Screw terminal (2 pin)	2	DEGSON DG350-3.5-02P-14-00A(H)			C2760668		https://www.lcsc.com	\$0.1706	\$0.3412
Screw terminal (3 pin)	1	DEGSON DG350-3.5-03P-14-00A(H)			C2760669		https://www.lcsc.com	\$0.1899	\$0.1899
								Total for PCB components	\$26.46
PCB	1							\$2.0000	\$2.0000
Stencil	1							\$7.0000	\$7.0000
								Total for PCB fabrication	\$9.00
XT60E-M connector	1				C3040679	XT60E-M.G.Y	https://www.lcsc.com	\$0.6084	\$0.6084
Handle connector pins	5				C17283940	1460-9-00-80-00-00-03-0	https://www.lcsc.com	\$0.3633	\$1.8165
Screws M2.5x6	4			For securing PCB and XT60 connector					
Screws M2.5x10	4			For securing enclosure lid and standby sense plate					
Nuts M2.5	2			For securing standby sense plate					
PCB standoffs L=16	2			For securing PCB and enclosure lid			https://eu.mouser.com		
Threaded inserts for plastic M2.5	4								
Ring terminal M2.5	1			For connecting standby sense wire to sense plate					
Enclosure 3D print	1							\$5.0000	\$5.0000
								Total for others	\$7.42
T245 handle	1						https://www.aliexpress.com	\$15.0000	\$15.0000
C245 tip	1							\$5.0000	\$5.0000
								Total for handle+tip	\$20.00
								Total	\$62.88